Which Tail, and Probability of What?

Unit 3 · Topic 3.6 · TODAY'S PROBLEM

Suppose a city's earlier planning model says 55% of households support expanded evening bus service. A researcher suspects support has *fallen*. From the city's 20,000 households, a simple random sample of 200 finds 98 that support expansion, so $\hat{p} = 0.49$. Conditions for a one-sample z-test are met.

Priya: "For $H_0 : p = 0.55$ versus $H_a : p < 0.55$, use the left-tail p-value, about 0.044. If support is still 55%, about 4.4% of random samples of 200 would produce a sample proportion of 0.49 or lower."
Noah: "Any difference from 0.55 counts, so double the tail to about 0.088. That means there is an 8.8% chance that the actual support proportion differs from 0.55."

Sample counts

Group	Support	Other	Total
Households	98	102	200

FIRST TAKE (5 min, on your sheet)

Which p-value and interpretation answer whether support has fallen? Choose Priya or Noah and cite one phrase or number.

DOK 1 **(a)** State the null and alternative hypotheses and identify the tail direction required by the research question.

DOK 2 **(b)** Compute the one-sample proportion z-statistic and its one-sided p-value. Also compute the two-sided p-value Noah would obtain.

DOK 3 **(c)** Choose the warranted p-value and interpretation. Use H_a and $\hat{p} = 0.49$ to define the tail event and null assumption, then explain what Noah's doubled area answers and what his statement gets wrong.

Video + follow-along: `u6_lesson5_live.html` (scan the code)
Finish (a)–(c) after the video. Turn in your sheet.